1 Isomorphisms

We have now seen a variety of vector spaces. Some of these, however, are not essentially different. For instance, any singleton set $\{v\}$ can be given the structure of the zero vector space by defining $v + v = v$ and $\alpha v = v$ for all $\alpha v = v$ for all $\alpha \in F$. There is no real difference, then, between any two vector spaces $\{v\}$ and $\{w\}$; all we have done is label the only vector in the set by a different symbol. Algebraically, these are exactly the same. In general, two vector spaces are the same in this sense if

1. there is a 1-1 correspondence (i.e a bijective map) between the underlying sets, and
2. this 1-1 correspondence preserves vector addition and scalar multiplication; that is, it is linear.



Definition 1.1 (Isomorphism)

Let V and W be vector spaces over F. An *isomorphism from V to W* is a bijective (injective and subjective) linear map $T : V \to W$. If there exists an isomorphism from V to W, we say that V and W are *isomorphic* and write $V \cong W$.



Example 1.2

Define $T : \{v\} \to \{w\}$ by $Tv = w$. It is immediately verified that T is a bijection, and it is also linear since

$$\begin{aligned} T(\alpha v + v) &= T(v) \text{ (since } +v = v) \\ &= w \\ &= \alpha w + w \\ &= \alpha T v + T v. \end{aligned}$$

Thus, T is a isomorphism from $\{v\}$ to $\{w\}$.



Exercise 1.3

Show that $F^n \cong M^{n\times 1}(F)$. One can similarly show that $F^n \cong M^{1\times n}(F)$. That is, ordered n-tuples can equivalently be represented as either row vectors or column vectors.



Exercise 1.4

Show that the subspace $W = \{(x, y, 0) \subset F^3\}$ is isomorphic to F^2.



Proof:

Define the map $T : F^2 \to W$ by $(x, y) \in F^2 \to$(x, y, 0) $\in F^3$. This is clearly linear and subjective by construction. To check injectivity, if $f((x, y)) = (0, 0, 0) \in F^3$, then $(x, y, 0) = (0, 0, 0)$, so $x, y = 0$ and so $(x, y) = 0$, showing the kernel is trivial and therefore injectivity.





Exercise 1.5

Show that $\mathcal{P}_n(F)$ (the vector space of all polynomials with coefficients in F of degree at most n) is isomorphic to F^{n+1}.

Proof:

Define the map $T(a_0 + a_1x^1 + a_2x^2 + ... + a_nx^n) = (a_0, a_1, ..., a_{n+1})$. This is again clearly linear and surjective by construction (since it is linear in each coefficient of x^i, and then these map to entries in F^{n+1}). It is injective since if $T(p(x)) = \mathbf{0}$, then $(a_0, a_1, ..., a_n) = (0, ..., 0)$, so $a_i = 0$ for all i, so $p(x) = \mathbf{0}$ in $\mathcal{P}_n(F)$. So, this is an isomorphism as desired.

Exercise 1.6

Recall the "unusual" vector space V defined by $V = \mathbf{R}^+$ (the positive reals) with addition and multiplication defined by:

$$x + y := xy$$
$$kx := x^k.$$

Show that $V \cong \mathbf{R}$.

Proof:

Define T by $x \to \log(x)$ (denote the addition and multiplication by $+_V$, $\cdot_V$ for clarity). We have $T(\alpha \cdot_V x +_V y) = \log(x^\alpha \cdot y) = \log(x^\alpha) + \log(y) = \alpha \log(x) + \log y = \alpha T(x) + T(y)$, showing linearity as desired. This is surjective since log on $(0, \infty)$ is surjective on $(-\infty, \infty)$ (for positive x, take $\log(e^x)$, for negative x, take $\log(e^{-x})$). It is injective since log is monotonically strictly increasing and as such takes on $\log(x) = 0$ only at $x = 1$, the 0 in 1, so the kernel is trivial.

2 Classification of Finite-Dimensional Vector Spaces

Proposition 2.1

(a) For any vector space V, $\mathrm{id}_V : V \to V$ is an isomorphism. Thus, $V \cong V$ for every V.

(b) If $T : V \to W$ is an isomorphism, then $T^{-1} : W \to V$ is also an isomorphism. Thus, $V \cong W$ implies $W \cong V$.

(c) If $T_1 : U \to V$ and $T_2 : V \to W$ are isomorphism, then $T_2T_1 : U \to W$ is an isomorphism. Thus, $U \cong V$ and $V \cong W$ implies $U \cong W$.

Theorem 2.1 says that "isomorphic" is reflexive, symmetric, and transitive, and is therefore an equivalence relation on the collection of all vector spaces over F.

Exercise 2.2

Prove Theorem 2.1. We have previously proved that the identity map is linear and that the composition of linear maps is linear. You have most likely also proved in a previous course that the identity map is a bijection, the inverse of a bijection is a bijection, and that the composition of two bijections is a bijection (convince yourself of these facts if you need to). All that remains, then, is to prove that if $T : V \to W$ is a linear bijection, then $T^{-1} : W \to V$ is also linear.

Proof:

Consider $T^{-1}(\alpha w_1 + w_2)$ for $w_1, w_2 \in W$. Since T is surjective, there exist v_1, v_2 such that $Tv_1 = w_1$ and $Tv_2 = w_2$, such that $T^{-1}w_1 = v_1$, $T^{-1}w_2 = v_2$. Then, we see $T(\alpha v_1 + v_2) = \alpha Tv_1 + Tv_2 = \alpha w_1 + w_2$ by linearity. Then, we see $T^{-1}(T(\alpha v_1 + v_2)) = T^{-1}(\alpha w_1 + w_2)$, but

$T^{-1}(T(\alpha v_1+v_2)) = \alpha v_1+v_2 = \alpha T^{-1}(w_1)+T^{-1}(w_2)$, so we see $T^{-1}(\alpha w_1+w_2), \alpha T^{-1}(w_1)+T^{-1}(w_2)$, showing linearity.

If T_1, T_2 are isomorphism, then $T_2 \circ T_1$ is bijective and linear, showing it is an isomorphism from $U \to W$.

Exercise 2.3

Theorem 2.2 says that "isomorphic" is an equivalence relation on the collection of all vector spaces over F. Define the *isomorphism class*, $[V]$, of a vector space V to be the set of all vector spaces which are isomorphic to $V : [V] = \{W : W \cong V\}$. Show that:

(a) Every vector space V over F belongs to some isomorphism class.

(b) Any two isomorphism classes are disjoint or equal.

Proof:

This is exactly the proof as it is for more generally cosets. But, since $V \cong V$ for all V, $V \in [V]$. For part b, consider isomorphism classes $[V], [W]$ and X such that $X \in [V], X \in [W]$. Then, for every $U \in V$, $U \cong V$ and $V \cong X$, so $U \cong X$ by transitivity, and then $X \cong W$, so $U \cong W$ by transitivity, so $U \in [W]$. This shows $[V] \subset [W]$. The reverse direction follows in the same manner, so $[W] \subset [V]$, so $[V] = [W]$.

Proposition 2.4

Let V be a finite-dimensional vector space of dimension $\dim V = n$ and let $B = (v_1, ..., v_n)$ be an ordered basis of V. Define $T_b : F^n \to V$ to be the linear combination map $T(x_1, ..., x_n) = x_1v_1 + ... + x_nv_n$. T_B is an isomorphism.

Proof:

Surjectivity and linearity of the map are clear by construction (it is clear pointwise in each basis vector and we extend to the sequence normally). The kernel is trivial since B is a basis, meaning the only combination of the v_i equal to zero is when all the x_i are zero (symbolically, if $T(v) = 0$, $v = 0$) making the map injective. Therefore, this is an isomorphism.

Corollary 2.5

If V, W are finite-dimensional vector spaces with $\dim V = \dim W = n$, then $V \cong W$.

Proof:

Choose a basis B_1 for V (which we can do since we know the dimension is n, i.e there exists a basis of size n) and apply T_{B_1} to V to see $V \cong F^n$. Similarly, we see $W \cong F^n$. Then, by equivalence property relations, $V \cong F^n \cong W$, so $V \cong W$ as desired.

Classification of Finite-Dimensional Vector Spaces: We have just proved there is a single isomorphism class of vector spaces over F for each non-negative integer. If $\dim V = 0$, then $V \cong \{0\}$. If $\dim V = n \geq 1$, then $V \cong F^n$.
Recall for a map of finite sets $f : A \to B$ with $|A| = |B|$, injectivity *or* surjectivity implies bijectivity. The following result for linear maps between finite-dimensional vector spaces of the same dimension is analogous.



Proposition 2.6

If V and W are finite-dimensional vector spaces of the same dimension and $T : V \to W$ a linear map, then if T is injective or surjective, it is an isomorphism.



Proof:

If T is injective, then we see $T(V)$ is a subspace of W of dimension n. But then, we must have $T(V) = W$ since W of dimension n, making T surjective. Therefore, T is a bijection and therefore an isomorphism of V and W.
If T is surjective, T restricts to a bijection T' on $V' \subset V$ (by choosing exactly one representative of each element - does this require choice?) such that $(T')^{-1}(W) = V' \subset V$. Linearity of $(T')^{-1}$ makes V' a subspace of V, and then $V' \cong W$ and so has dimension n. Then, we must have $V' = V$ since V' has dimension n and V has dimension n, so this shows $(T')^{-1}$ is an isomorphism of V and W.





Exercise 2.7

Use the above proposition to prove given any polynomial q, there exists a polynomial p of the same degree such that $((x^2 + 5x + 7)p)'' = q$.



Proof:

If $\deg(q) = n$, then work in $\mathcal{P}_n(F)$. Consider the map $f : p(x) \to (((x^2+5x+7)p)'')$. This is clearly linear and a map from $\mathcal{P}_n(F) \to \mathcal{P}_n(F)$, since it is a composition of the multiplication by (x^2+5x+7) from $\mathcal{P}_n(F) \to \mathcal{P}_{n+2}(F)$ and derivative maps $\mathcal{P}_{n+2}(F) \to \mathcal{P}_{n+1}(F)$ and $\mathcal{P}_{n+1}(F) \to \mathcal{P}_n(F)$.
Now, consider p such that $((x^2+5x+7)p)'' = 0$. Integrating, we must have $(x^2+5x+7)p = c_1x+b$. But, if $p(x)$ has any nonzero terms, it has a highest-degree term c_ix^i which by the zero product in $F[x]$ multiplies to a term c_ix^{i+2} of degree ≥ 2 on the LHS that does not vanish. This is not compatible with the right-hand side. So, we must have $p = 0$, showing that f is injective. Then, by the proposition, f is surjective and we are done.

